

Problem (informal): On the pure two-qubit state space $\mathbb{CP}^3$ with the Fubini–Study metric, study the spectrum and time-evolution of a weighted Hodge–de Rham Laplacian (on appropriate forms) with a thermal weight derived from a two-qubit Hamiltonian. Interpret the low-lying modes and dynamics as describing how probability distributions over pure states thermalize and how distinguishability/entanglement structure behaves under these dynamics.

SECTION 1 – THE GEOMETRIC SETUP OF THE PROBLEM.

We start by defining our Manifold. Let $\mathcal{M} = \mathbb{CP}^3$, the set obtained by taking $\mathbb{C}^4 - \{0\}$ modulo the equivalence relation:

$$z \sim w \Leftrightarrow z = \lambda w, \lambda \in \mathbb{C}^*$$

Which is the set of complex lines in $\mathbb{C}^4$ through the origin, of course under the usual projection topology that allows us to define the following chart

The standard atlas. For each $k = 0, 1, 2, 3$ define the open set

$$U_k = \{[z_0 : z_1 : z_2 : z_3] \in \mathbb{CP}^3 : z_k \neq 0\}.$$

So on each U_k , we're sure that at least the $z_k \neq 0$, which allows us to define the map.

$$\begin{cases} \phi_k : \mathcal{M} \rightarrow \mathbb{C}^3 \\ p = [z_i] \rightarrow \phi_k(p) = (w_i)_{i \in \{0,1,2,3\} - \{k\}} \end{cases} \quad \text{with } w_i = \frac{z_i}{z_k}$$

It is important to note that $\mathcal{M}$ isn't an ordinary manifold, it's a *Kähler* manifold:

A **Kähler manifold** is a complex manifold whose metric is simultaneously Riemannian, Hermitian, and symplectic, all in a compatible way; the **Kähler form** is the 2-form that encodes this metric plus the complex structure, and its being closed is what makes the manifold “Kähler.”

A Riemannian metric g on M is **Hermitian** if it is compatible with J , meaning

$$g(JX, JY) = g(X, Y) \text{ for all tangent vectors } X, Y.$$

The pair (g, J) gives an almost Hermitian manifold. From this data you define the associated 2-form

$$\omega(X, Y) = g(JX, Y), X, Y \in TM.$$

- ω is automatically nondegenerate and of type $(1, 1)$ (in complex terms), so it is a candidate symplectic form. [wikipedia+1](#)

The manifold is **Kähler** if, in addition, this 2-form is closed:

$$d\omega = 0.$$

So a Kähler manifold is equivalently:

- A complex manifold with a Hermitian metric whose associated 2-form ω is closed.
- Or a symplectic manifold (M, ω) equipped with an integrable complex structure J such that $\omega(\cdot, J\cdot)$ is a Riemannian metric

This triple (g, J, ω) is the Kähler structure, and $\mathbb{CP}^n$ with the Fubini–Study metric is the standard metric, which we’ll be taking for this problem.

The round metric on $\mathbb{S}^{2n+1}$ descends to a homogeneous and isotropic Riemannian metric on $\mathbb{CP}^{n+1}$, called the Fubini-Study metric.

On a complex manifold with local holomorphic coordinates $z_1, \dots, z_n$, the Kähler metric g is encoded by a function $\rho(z, \bar{z})$ such that

$$\omega = \frac{i}{2} \partial \bar{\partial} \rho$$

is the Kähler 2-form, and its components $g_{i\bar{j}}$ are

$$g_{i\bar{j}} = \frac{\partial^2 \rho}{\partial z_i \partial \bar{z}_j}.$$

This function ρ is called a **Kähler potential** for ω (and for the metric g).

On the affine chart $Z_0 \neq 0$ with $w_i = Z_i/Z_0$ for $i = 1, 2, 3$:

- The Kähler potential is

$$\rho(w, \bar{w}) = \log(1 + |w_1|^2 + |w_2|^2 + |w_3|^2).$$

- The metric components are

$$g_{i\bar{j}}(w) = \frac{\partial^2 \rho}{\partial w_i \partial \bar{w}_j}.$$

- The Kähler form is

$$\omega = \frac{i}{2} \sum_{i,j} g_{i\bar{j}} dw_i \wedge d\bar{w}_j.$$

Chart U_0 : $Z_0 \neq 0$

- Coordinates: $w_1 = Z_1/Z_0, w_2 = Z_2/Z_0, w_3 = Z_3/Z_0$.
- Potential: $\rho_0 = \log(1 + |w_1|^2 + |w_2|^2 + |w_3|^2)$.
- Metric: $g_{ij}^{(0)}(w) = \frac{\delta_{ij}}{1 + |w_1|^2 + |w_2|^2 + |w_3|^2} - \frac{\bar{w}_i w_j}{(1 + |w_1|^2 + |w_2|^2 + |w_3|^2)^2}, i, j = 1, 2, 3$.

Chart U_1 : $Z_1 \neq 0$

- Coordinates: $u_0 = Z_0/Z_1, u_2 = Z_2/Z_1, u_3 = Z_3/Z_1$.
- Potential: $\rho_1 = \log(1 + |u_0|^2 + |u_2|^2 + |u_3|^2)$.
- Metric: $g_{pq}^{(1)}(u) = \frac{\delta_{pq}}{1 + |u_0|^2 + |u_2|^2 + |u_3|^2} - \frac{\bar{u}_p u_q}{(1 + |u_0|^2 + |u_2|^2 + |u_3|^2)^2}, p, q \in \{0, 2, 3\}$

And so on and so forth...

now that we set up our Metric, we'd want to work with its volume form.

Recall The Kähler form is

$$\omega_{\text{FS}} = \frac{i}{2} \sum_{i,j=1}^3 g_{ij}(w) dw_i \wedge d\bar{w}_j.$$

On any **oriented Riemannian manifold** (M, g) , the volume form is the unique top-degree form dvol_g such that for any oriented orthonormal basis $(e_1, \dots, e_n)$ of $T_p M$ you have

$$\text{dvol}_g(e_1, \dots, e_n) = 1.$$

In local coordinates, this condition forces

$$\text{dvol}_g = \sqrt{\det(g_{ij})} dx^1 \wedge \dots \wedge dx^n.$$

So “volume form = $\sqrt{\det g}$ times coordinate wedge” is not a choice; it's the unique form that makes the integral of 1 equal the geometric volume.

On a **Kähler manifold** of complex dimension n :

- You have a Hermitian metric $g_{i\bar{j}}$ and its associated Kähler form

$$\omega = \frac{i}{2} \sum_{i,j} g_{i\bar{j}} dz_i \wedge d\bar{z}_j.$$

- In an orthonormal unitary frame, ω looks like

$$\omega = \sum_{k=1}^n e^k \wedge f^k$$

where $\{e^k, f^k\}$ is an orthonormal real coframe (think $\Re dz_k, \Im dz_k$). [wikipedia+1](#)

Then the wedge power satisfies

$$\omega^{\wedge n} = n! \, e^1 \wedge f^1 \wedge \cdots \wedge e^n \wedge f^n,$$

which is exactly $n!$ times the Riemannian volume form. [ncatlab+1](#)

So the *definition* becomes

$$\mathrm{dvol}_g = \frac{1}{n!} \, \omega^{\wedge n}$$

In these coordinates, that becomes

$$\mathrm{dvol}_{\mathrm{FS}} = \frac{1}{3!} \left(\frac{i}{2}\right)^3 \det(g_{i\bar{j}}(w)) \, dw_1 \wedge d\bar{w}_1 \wedge dw_2 \wedge d\bar{w}_2 \wedge dw_3 \wedge d\bar{w}_3.$$

Tangent vectors as infinitesimal state changes

Take a normalized state $|z\rangle \in \mathbb{C}^4$ (Hilbert space), and its ray $[z] \in \mathcal{M}$.

By definition, $[z]$ is the equivalence class of all nonzero vectors proportional to $|z\rangle$, i.e. $\lambda |z\rangle$ with $\lambda \in \mathbb{C}^\times$; physically that's “same state up to global phase and normalization.”

To see what a tangent vector at $[z]$ means:

- Think of the quotient map $\pi: \mathbb{C}^4 \setminus \{0\} \rightarrow \mathcal{M}$, $\pi(|\psi\rangle) = [\psi]$. [wikipedia+1](#)
- A tangent vector $X \in T_{[z]}\mathbb{CP}^3$ can be represented by a derivative of a curve of states

$$t \mapsto |\psi(t)\rangle, |\psi(0)\rangle = |z\rangle,$$

modulo changes along the fiber (i.e. multiplying by a phase). [idv.sinica+1](#)

Concretely, one standard description is:

$$T_{[z]}\mathcal{M} \cong \frac{\{v \in \mathbb{C}^4: \langle z | v \rangle = 0\}}{\text{phase directions}}.$$

Intuitively:

- A vector v orthogonal to $|z\rangle$ describes an infinitesimal motion in Hilbert space that changes the *ray* nontrivially.
- Adding a term proportional to $|z\rangle$ (e.g. $i\alpha |z\rangle$) only changes the global phase or normalization, i.e. it moves along the fiber of π and does **not** change the point $[z]$ in $\mathcal{M}$

If a small Hamiltonian H generates Schrödinger evolution,

$$|\psi(t)\rangle = e^{-itH} |z\rangle,$$

then at $t = 0$ the derivative is

$$\frac{d}{dt} |_{t=0} \psi(t) = -iH |z\rangle.$$

We decompose $-iH |z\rangle$ into a part parallel to $|z\rangle$ (global phase and normalization) and a part orthogonal to $|z\rangle$. The orthogonal part is precisely the tangent vector in $T_{[z]}\mathbb{CP}^3$, i.e. the actual **direction of state change modulo global phase**

1-form A as a quantum current / cost functional

A 1-form $A \in \Omega^1(\mathbb{CP}^3)$ assigns to each point $[z]$ a linear functional

$$A_{[z]}: T_{[z]}\mathbb{CP}^3 \rightarrow \mathbb{C}.$$

So for a tangent vector $X \in T_{[z]}\mathbb{CP}^3$, the value $A(X)$ is:

- A scalar depending linearly on the direction X .
- Hence a **directional cost**, “current,” or “instantaneous rate” associated with moving in that direction on the state space. [wikipedia](#)+1

Interpreted physically, we can choose A so that:

- $A(X)$ is the instantaneous **work** per unit time to follow the change encoded by X .
- Or the instantaneous **phase** change per unit time (like a Berry connection acting on state-space paths).
- Or an **information-theoretic cost**, e.g. rate of change of an expectation or entropy functional along X

given a small change in state $[z] \rightarrow [z + \delta z]$ represented by X , $A(X)$ quantifies how “expensive” this change is according to the chosen functional A . In FE terms, think of A as a linear form on the tangent space that you plug your vector field X into.

Section 2 – Weighted Hodge operator on 1-forms

Pick a **Hermitian** K with two features:

1. Non-degenerate spectrum (to avoid trivial symmetries).
2. A block structure that you can interpret physically—e.g. a pair of coupled qubits, or a 4-level atom with two near-degenerate subspaces.

For instance:

$$K = \begin{pmatrix} \epsilon & 0 & 0 & 0 \\ 0 & -\epsilon & g & 0 \\ 0 & g & 0 & 0 \\ 0 & 0 & 0 & \Delta \end{pmatrix}$$

with ϵ, Δ encoding energy splittings and g a coupling between levels 2 and 3.

Then define:

$$f([z]) = \langle z | K | z \rangle, \quad |z\rangle \in \mathbb{C}^4, \quad \|z\| = 1, \quad [z] \in \mathcal{M},$$

and the 1-form

$$A = df$$

on $\mathbb{CP}^3$. As we already noted, $A(X)$ for $X \in T_{[z]}\mathcal{M}$ is the instantaneous rate of change of the expectation of K along that direction of state

Now define the **geometric Hamiltonian** for your project as an operator on scalar functions or 1-forms on $\mathbb{CP}^3$:

- Scalar version:

$$H_{\text{scalar}} = -\Delta_{\text{FS}} + \alpha f([z]),$$

with α a coupling parameter.

- 1-form version (closer to your FEEM on 1-forms):

$$H_{\text{forms}} = \Delta_{\text{Hodge}} + \alpha f([z]) \text{Id},$$

Define the **thermal weight** at inverse temperature $\beta > 0$:

$$d\mu_\beta = e^{-\beta E([z])} d\text{vol}_{\text{FS}}.$$

We are defining this measure because you need a **geometric, physically meaningful L^2 inner product** on fields (functions or forms) that matches both the manifold's metric and the physics we're modeling (mass, potential, weights).

- The volume form $d\text{vol}_g$ turns the pointwise inner product coming from the Riemannian metric into an L^2 scalar product by integration.
- A weight (like an OU-type factor or geodesic trap) modifies the measure so that the operator you care about (Hodge–de Rham, OU/Witten Laplacian, Hamiltonian with potential) becomes self-adjoint on the corresponding weighted L^2 space.

At each p , g_p gives an inner product on $T_p M$ and hence on $T_p^* M$

$T_p^* M$ is the dual of $T_p M$: its elements are linear maps $T_p M \rightarrow \mathbb{R}$. Since g_p is non-degenerate, there is a canonical isomorphism (the flat/sharp map):

$$b: T_p M \rightarrow T_p^* M, \quad v \mapsto v^b := g_p(v, \cdot).$$

And Its inverse is $\sharp = b^{-1}$

This lets you transfer the inner product from $T_p M$ to $T_p^* M$: given two covectors $\alpha, \beta \in T_p^* M$, define $\langle \alpha, \beta \rangle_{T_p^* M} := g_p(\alpha^\sharp, \beta^\sharp)$. In coordinates, if (g_{ij}) is the inverse metric matrix, then

$$\langle dx^i, dx^j \rangle_{T_p^* M} = g^{ij}(p)$$

- In coordinates with g_{ij} and inverse g^{ij} , if $\alpha = \alpha_i dx^i$, $\beta = \beta_j dx^j$, then

$$\langle \alpha, \beta \rangle_p = g^{ij}(p) \alpha_i(p) \beta_j(p).$$

This gives you a scalar at each point, exactly like the dot product of vectors but using the metric to raise indices.

L^2 inner product on 1-forms

To get the global “dot product” you use in your weak formulations and mass matrices, you integrate the pointwise inner product against the chosen measure (usually $d\text{vol}_g$).

- The L^2 inner product of 1-forms is

$$\langle \alpha, \beta \rangle_{L^2} = \int_M \langle \alpha, \beta \rangle_p d\text{vol}_g(p).$$

In local coordinates this is

$$\langle \alpha, \beta \rangle_{L^2} = \int_M g^{ij}(x) \alpha_i(x) \beta_j(x) \sqrt{\det g(x)} dx^1 \cdots dx^n$$

Then using our Fubini-Study metric on $\mathcal{M}$ Define the weighted L^2 inner product on 1-forms:

$$\langle A, B \rangle_{L^2(\mu_\beta)} = \int_{\mathcal{M}} \langle A, B \rangle_{FS} e^{-\beta E([z])} d\text{vol}_{FS}.$$

On $\mathbb{CP}^3$ with the Fubini-Study metric g_{FS} , the pointwise inner product $\langle A, B \rangle_{FS}([z])$ of smooth 1-forms A, B is a smooth function on $\mathbb{CP}^3$.

The energy function $E([z])$ is smooth and real-valued, and the weight $e^{-\beta E([z])}$ is positive and smooth for all $[z]$ and $\beta \in \mathbb{R}$. The Fubini-Study volume form $d\text{vol}_{FS}$ is a smooth positive measure on the compact manifold $\mathbb{CP}^3$.

Putting these together, the integrand

$$\langle A, B \rangle_{FS}([z]) e^{-\beta E([z])}$$

is a smooth function, and the integral

$$\langle A, B \rangle_{L^2(\mu_\beta)} = \int_{\mathbb{CP}^3} \langle A, B \rangle_{FS}([z]) e^{-\beta E([z])} \mathrm{dvol}_{FS}([z])$$

is finite for all smooth 1-forms A, B because $\mathbb{CP}^3$ is compact and the weight $e^{-\beta E}$ is bounded above and below by positive constants.

Bilinearity, symmetry, and positivity

For smooth 1-forms A, B, C and scalars λ, μ , linearity of the integral and of $\langle \cdot, \cdot \rangle_{FS}$ gives:

- $\langle \lambda A + \mu B, C \rangle_{L^2(\mu_\beta)} = \lambda \langle A, C \rangle_{L^2(\mu_\beta)} + \mu \langle B, C \rangle_{L^2(\mu_\beta)}.$
- $\langle A, B \rangle_{L^2(\mu_\beta)} = \langle B, A \rangle_{L^2(\mu_\beta)}$ because $\langle A, B \rangle_{FS}$ is symmetric pointwise.
- $\langle A, A \rangle_{L^2(\mu_\beta)} = \int_{\mathbb{CP}^3} \langle A, A \rangle_{FS} e^{-\beta E} \mathrm{dvol}_{FS} \geq 0$, and it is zero if and only if $A = 0$ as a 1-form, since $\langle A, A \rangle_{FS}([z]) > 0$ wherever $A([z]) \neq 0$ and the weight and volume form are positive.

Thus $\langle \cdot, \cdot \rangle_{L^2(\mu_\beta)}$ is an inner product on the space of smooth 1-forms $\Omega^1(\mathbb{CP}^3)$.

Hilbert space via completion

The inner product defines a norm

$$\|A\|_{L^2(\mu_\beta)}^2 = \int_{\mathbb{CP}^3} \langle A, A \rangle_{FS} e^{-\beta E} \mathrm{dvol}_{FS},$$

which is equivalent to the standard L^2 norm defined with respect to dvol_{FS} alone, because the weight $e^{-\beta E}$ is bound above and away from zero on the compact manifold. Therefore, the metric topology induced by $\|\cdot\|_{L^2(\mu_\beta)}$ is the same as the usual L^2 topology on 1-forms, and the completion of $\Omega^1(\mathbb{CP}^3)$ under this norm is a Hilbert space $L^2(\mathbb{CP}^3, \Lambda^1 T^* \mathbb{CP}^3; \mu_\beta)$.

Recall the Hodge–de Rham Laplacian on 1-forms for the unweighted metric:

$$\Delta_1 A = d\delta A + \delta dA,$$

where δ is the codifferential defined via the Hodge star of g_{FS} .

Changing from $\mathbf{dvol}_{FS}$ to $\mathbf{d}\mu_\beta = e^{-\beta E} \mathbf{dvol}_{FS}$ modifies the adjoint of d by adding a **drift term** proportional to ∇E .

Let $\mu_\beta = e^{-\beta E} \mathbf{dvol}_{FS}$ on $\mathbb{CP}^3$, for smooth forms α and ω , the adjoint δ_β of d with respect to the weighted inner product $\langle \cdot, \cdot \rangle_{L^2(\mu_\beta)}$ satisfies

$$\int_{\mathcal{M}} \langle d\omega, \alpha \rangle_{FS} e^{-\beta E} \mathbf{dvol}_{FS} = \int_{\mathcal{M}} \langle \omega, \delta_\beta \alpha \rangle_{FS} e^{-\beta E} \mathbf{dvol}_{FS}.$$

To relate δ_β to the usual codifferential δ (adjoint with respect to $\mathbf{dvol}_{FS}$), write

$$\int_{\mathcal{M}} \langle d\omega, \alpha \rangle_{FS} e^{-\beta E} \mathbf{dvol}_{FS} = \int_{\mathcal{M}} \langle d\omega, e^{-\beta E} \alpha \rangle_{FS} \mathbf{dvol}_{FS}.$$

By definition of δ (unweighted adjoint of d) we have

$$\int \langle d\omega, e^{-\beta E} \alpha \rangle_{FS} \mathbf{dvol}_{FS} = \int \langle \omega, \delta(e^{-\beta E} \alpha) \rangle_{FS} \mathbf{dvol}_{FS}.$$

Using the Leibniz rule and the fact that δ is first order and metric-dependent, the unweighted codifferential applied to $e^{-\beta E} \alpha$ splits into

$$\delta(e^{-\beta E} \alpha) = e^{-\beta E} \delta \alpha + (\text{drift term involving } \nabla E).$$

More precisely, in the standard Witten/OU setting one gets a contraction with the gradient of E .

Matching the integrals then gives the formal expression

$$\delta_\beta \alpha = \delta \alpha + \beta \iota_{\nabla E} \alpha,$$

where ∇E is the gradient of E with respect to g_{FS} and $\iota_{\nabla E}$ denotes interior product with ∇E .

This $\beta \iota_{\nabla E}$ is the **drift operator** associated with the weight $e^{-\beta E}$: it is the first-order term that appears in the weighted Hodge Laplacian

$$\Delta_\beta = d \delta_\beta + \delta_\beta d = \Delta + \beta (d \iota_{\nabla E} + \iota_{\nabla E} d),$$

so relative to the unweighted Laplacian Δ you have an extra drift generated by the vector field ∇E .

Define a **weighted Hodge operator** on 1-forms:

$$\mathcal{H}_\beta A := \Delta_\beta A + V([z]) A = \Delta_1 A + \beta \mathcal{L}_{\nabla E} A + V([z]) A$$

SECTION 3: SOLVING THE EIGENVALUE PROBLEM: